SOFTWARE TESTING PRACTICAL REPORT

PR-15 | Selenium WebDriver Automation Testing

Particular	Details
Student Name	Parth Fegade
Enrollment Number	_____
Course / Semester	BCA – Semester 3
Practical	PR-15
Application Under Test	SauceDemo (https://www.saucedemo.com/)
Tools Used	Python, Selenium WebDriver, Google Chrome, PyCharm

Submitted as part of the Software Testing practical work.

1. Practical Overview

This practical demonstrates automated authentication testing on the SauceDemo website using Python and Selenium WebDriver. The test suite covers successful login, rejection of invalid credentials, validation of empty credentials, assertions, error-message verification, exception handling, and logout. The screenshots supplied with the practical are included as execution evidence.

2. Aim

To create and execute automated login tests using Selenium WebDriver with Python, validate authentication behavior for valid and invalid inputs, use assertions and exception handling, and document the observed results with screenshots.

3. Problem Statement

Create an automated Selenium login-testing script that performs valid login, invalid-password testing, invalid-username testing, empty-credential testing, authentication assertions, error-message validation, exception handling, and logout.

4. Software and Requirements

Parameter	Details
Project Name	Automated Login Testing
Application / Software	SauceDemo
Module	Authentication / Login
SDLC Model	Agile
Programming Language	Python 3.x
IDE / Tool	PyCharm
Automation Tool	Selenium WebDriver
Browser	Google Chrome
WebDriver	Selenium-managed Chrome WebDriver
Target URL	https://www.saucedemo.com/
Input / Test Data	Valid and invalid login credentials

5. Test Coverage

Test Area	Expected Behavior / Verification
Browser launch	Chrome opens under Selenium control.
Website navigation	SauceDemo login page loads.
Valid authentication	Valid credentials open the inventory page.
Invalid password	An authentication error is displayed.
Invalid username	An authentication error is displayed.
Empty credentials	A required-username validation message is displayed.
Assertions	URL and Products page title are checked.
Exception handling	Relevant Selenium exceptions are caught and reported.
Logout	The user returns to the login page.
Evidence capture	Screenshots document browser states and console output.

6. Login Automation Scenarios

Scenario ID	Scenario	Input	Expected Result
TC-01	Valid Login	standard_user / secret_sauce	Login succeeds; inventory page appears.
TC-02	Invalid Password	standard_user / wrong_password	Authentication error appears.
TC-03	Invalid Username	wrong_user / secret_sauce	Authentication error appears.
TC-04	Empty Credentials	Username and password blank	Required-field message appears.
TC-05	Logout	Valid login, then Logout	Login page appears again.

7. Procedure

1. Open PyCharm and create a Python project named SoftwareTesting.
2. Create a Python file named tutorial15.py.
3. Install Selenium in the project environment using: pip install selenium
4. Import webdriver, By, WebDriverWait, expected_conditions, and the required exception classes.
5. Launch Chrome with webdriver.Chrome() and maximize the browser window.
6. Open https://www.saucedemo.com/ and locate the username, password, and Login controls.
7. Run TC-01 with standard_user and secret_sauce; verify that the URL contains inventory and the page title is Products.
8. Run TC-02 using standard_user and wrong_password; verify that the authentication error appears.
9. Run TC-03 using wrong_user and secret_sauce; verify that the authentication error appears.
10. Run TC-04 with both fields blank; verify the Username is required message.
11. Run TC-05 by logging in, opening the navigation menu, selecting Logout, and verifying the login page.
12. Capture screenshots of the relevant browser states and PyCharm console output.

8. Web Element Identification

Element Name	Locator Type	Locator Value	Purpose
Username	ID	user-name	Enter username.
Password	ID	password	Enter password.
Login Button	ID	login-button	Submit login form.
Error Message	CSS Selector	h3[data-test='error']	Validate authentication or input error.
Menu Button	ID	react-burger-menu-btn	Open navigation menu.
Logout Button	ID	logout_sidebar_link	Log out of the application.

9. Assertions and Exception Handling

Assertion Validation

Assertion	Expected Result	Validation Purpose
URL contains inventory	Inventory URL is reached	Confirms successful navigation after login.
Page title equals Products	Products	Confirms the inventory page loaded.
Invalid login error text	Credentials mismatch message	Confirms invalid credentials are rejected.
Empty username error text	Username is required	Confirms empty-input validation.
Logout returns to login	Login form is available	Confirms logout behavior.

Exception Handling Report

Exception	Possible Cause	Handling / Remarks
NoSuchElementException	A required element cannot be located.	Caught and reported by the test code.
TimeoutException	An element or state does not appear before the wait expires.	Caught and reported by the test code.
ElementNotInteractableException	An element cannot be interacted with.	Caught and reported by the test code.
StaleElementReferenceException	The page DOM changes after an element is located.	Caught and reported by the test code.
AssertionError	Actual behavior differs from expected behavior.	Marks the relevant test as failed.

10. Test Case Execution Summary

Metric	Value / Interpretation
Total Test Cases	5
Test Cases in Scope	TC-01 through TC-05
Expected Successful Scenarios	Valid login and logout
Expected Rejection Scenarios	Invalid password, invalid username, empty credentials
Execution Evidence	Screenshots and console output supplied
Result Note	Console screenshot visibly reports TC-01 through TC-05 as PASS; verify the complete run locally before submission.

11. Observations

- The SauceDemo login page was opened in Chrome under Selenium control.
- The supplied screenshots show the inventory page after a successful login flow.
- The supplied invalid-credentials screenshot displays the expected authentication error.
- The navigation menu is visible in the supplied inventory-page screenshot, supporting the logout workflow.
- The PyCharm console screenshot lists TC-01 Valid Login, TC-02 Invalid Password, TC-03 Invalid Username, TC-04 Empty Credentials, and TC-05 Logout as PASS.
- A Chrome Password Manager warning appears in one screenshot. It is generated by the browser and is not a SauceDemo application message.

12. Execution Evidence (Screenshots)

The following screenshots were provided and are included as visual evidence. Captions describe the visible state; they do not replace rerunning the practical on the local machine.

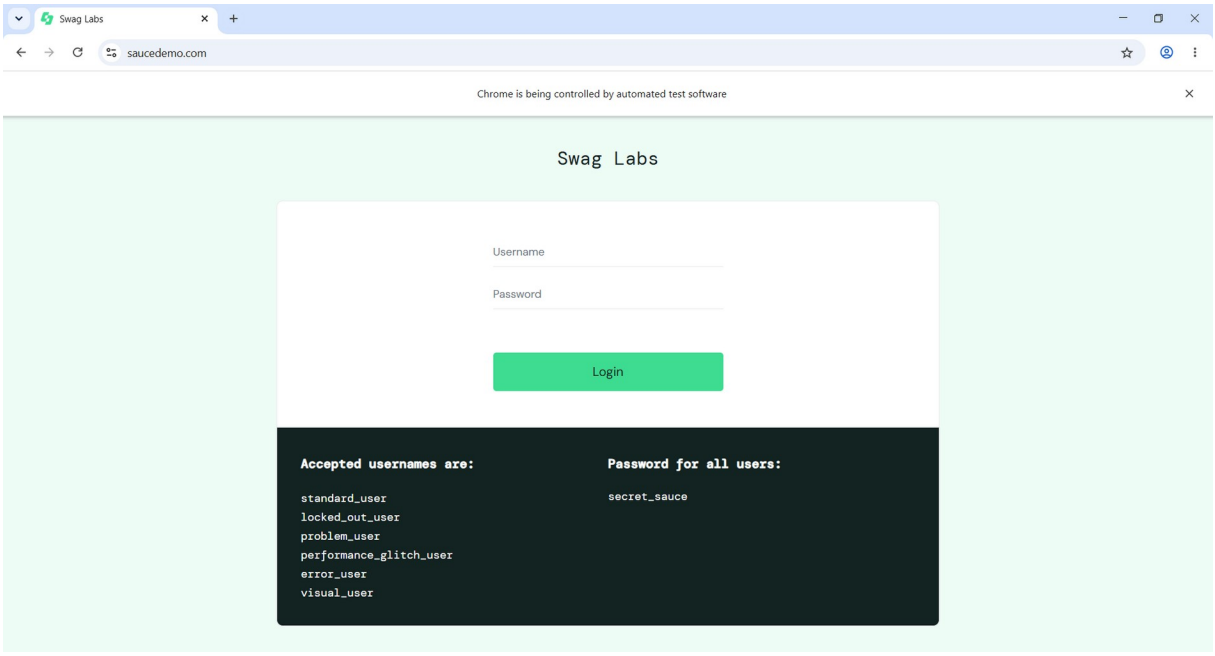


Figure 1. SauceDemo login page before entering credentials.

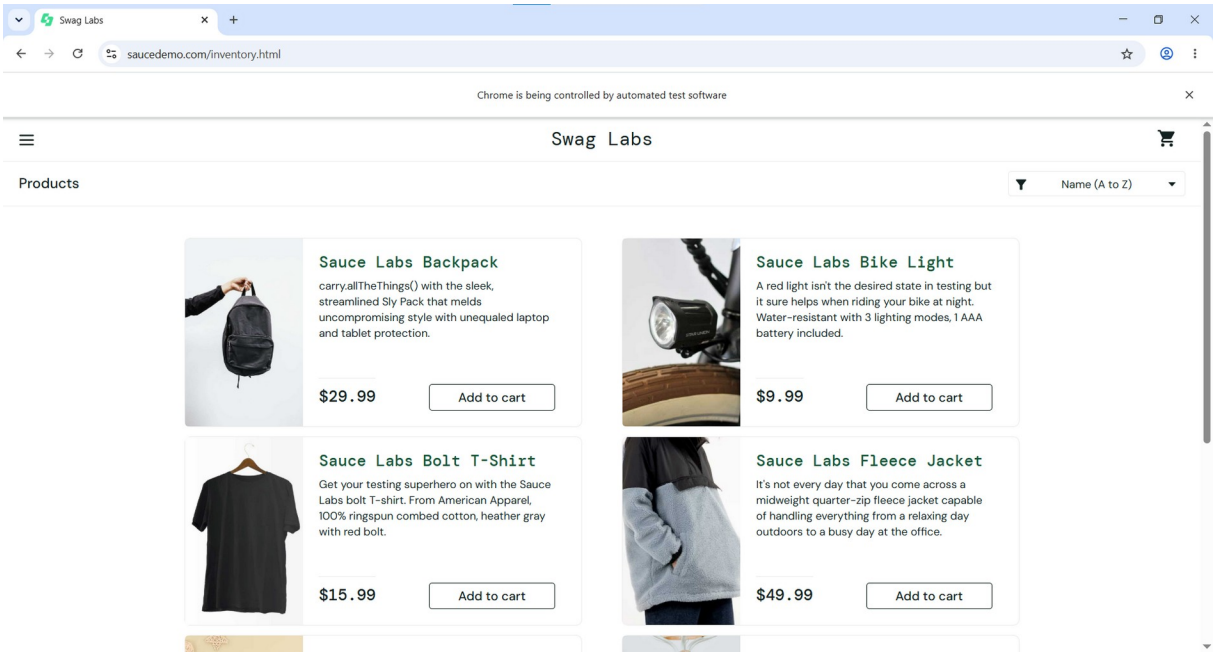


Figure 2. SauceDemo inventory page. A Chrome Password Manager warning is visible; this is a browser notification, not part of the application.

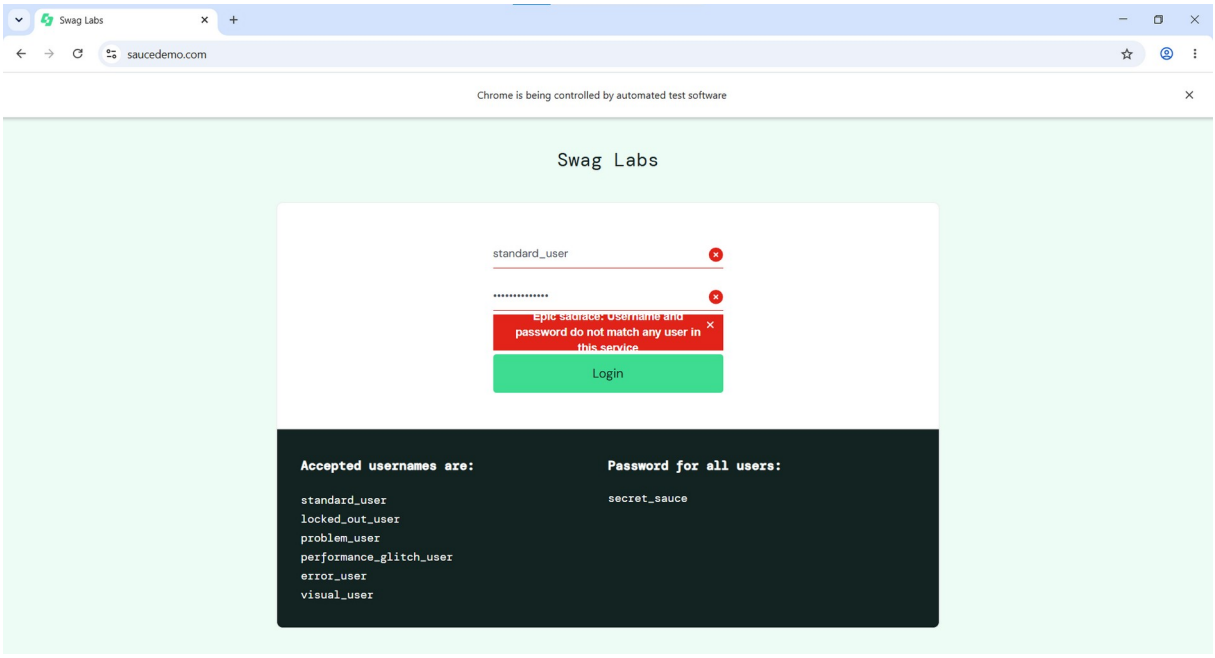


Figure 3. PyCharm source code and console output from a Selenium run.

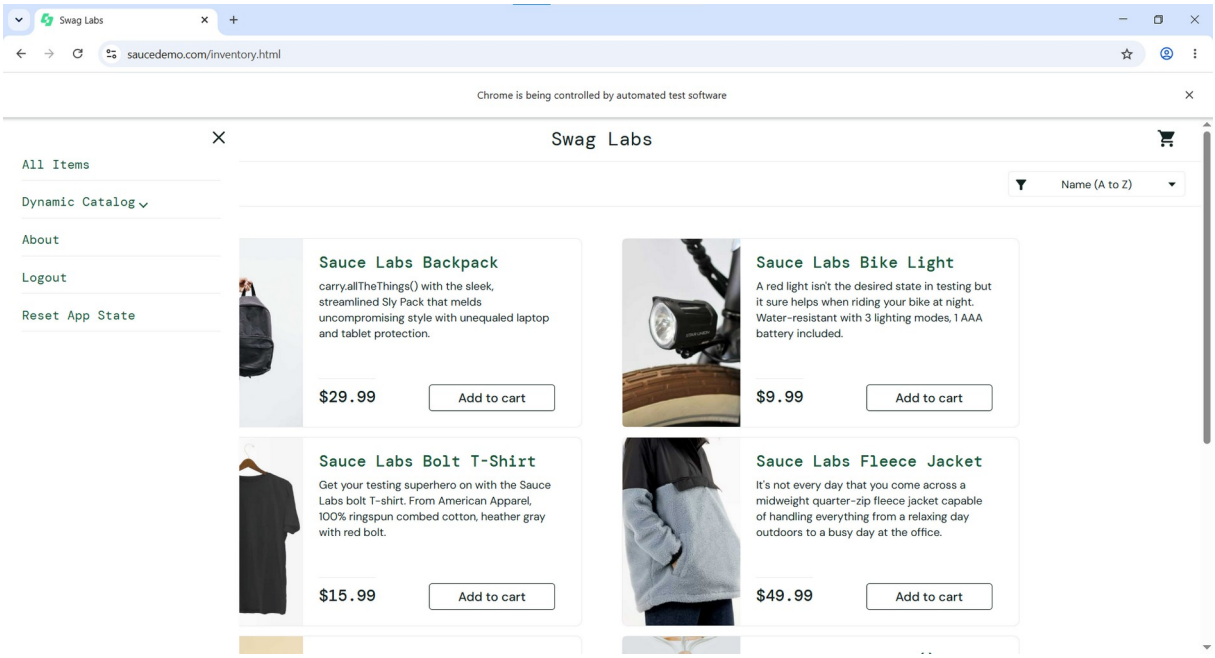


Figure 4. SauceDemo login page with invalid-credentials error displayed.

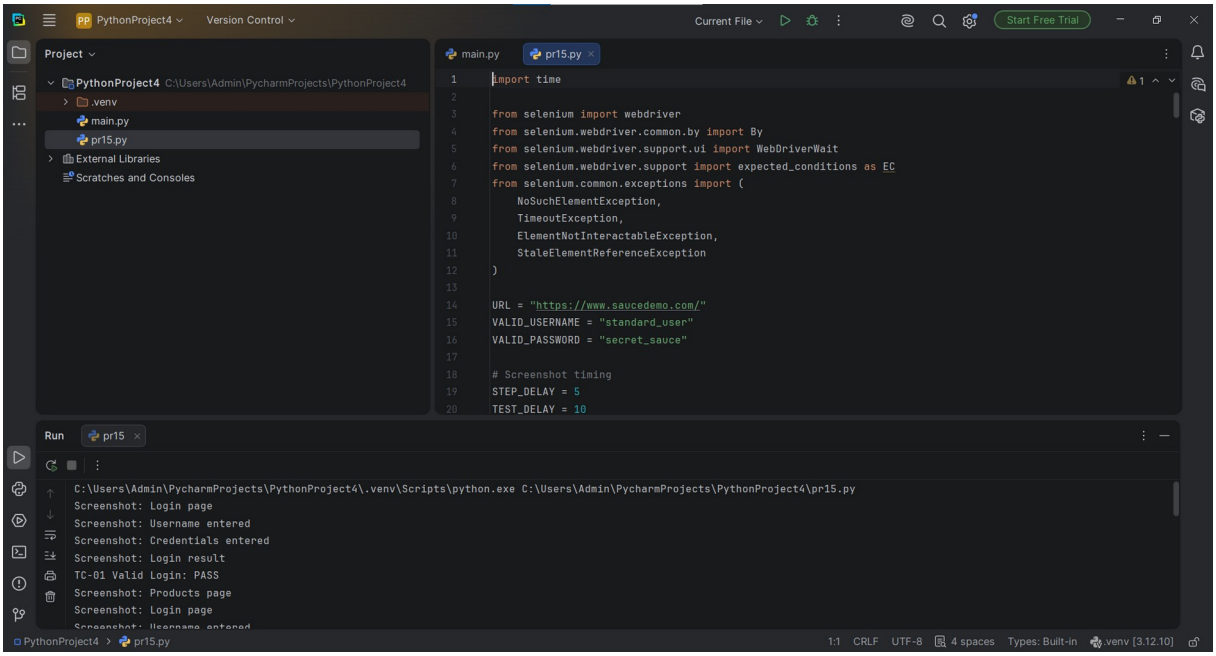


Figure 5. SauceDemo inventory page with the navigation menu open.

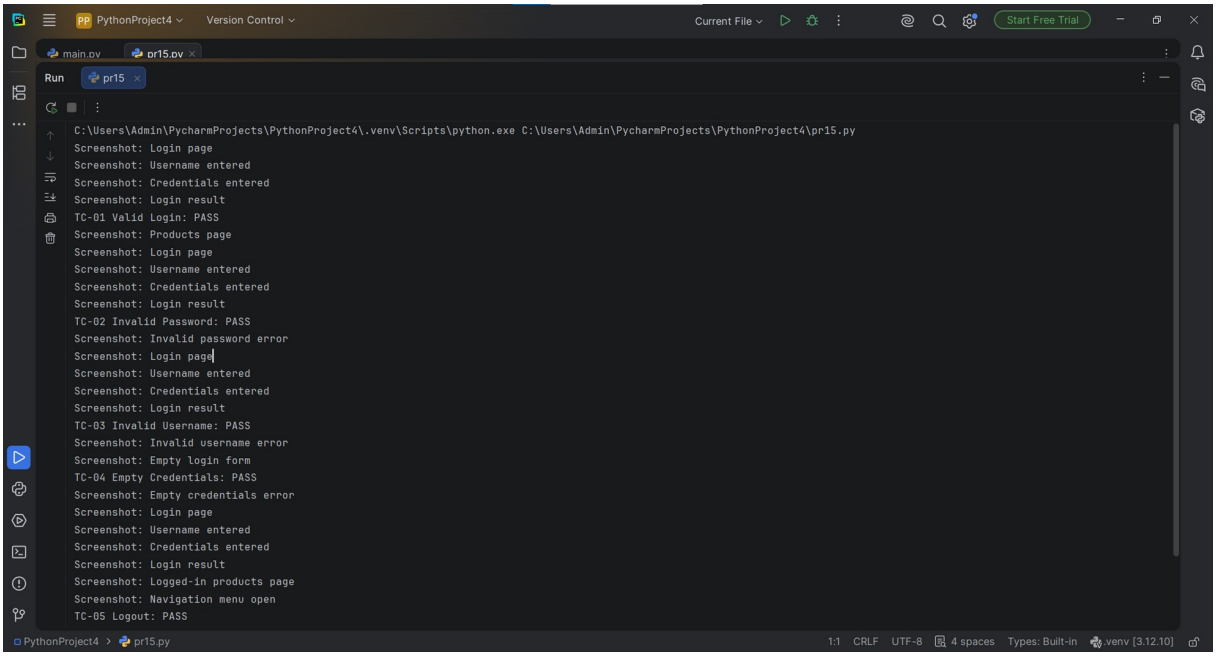


Figure 6. PyCharm console output showing the login test cases and logout test result.

13. Result

The provided execution evidence supports that the SauceDemo login and inventory workflow was automated with Selenium WebDriver. The visible PyCharm console output reports the five required scenarios as PASS. The report records these outcomes based on the supplied screenshots; the student should retain the complete console output from a fresh local execution for final verification.

14. Conclusion

This practical demonstrates how Selenium WebDriver can automate login testing for a web application. It covers positive and negative authentication scenarios, input validation, assertions, exception handling, and logout. Explicit waits help synchronize the automation with browser elements, while assertions compare actual application behavior with expected results. The screenshots provide supporting evidence of the login page, inventory page, error state, navigation menu, and test console.

15. Questionnaire

Q1. What is authentication testing?

Answer: Authentication testing verifies that valid users are allowed to access an application and invalid users are rejected.

Q2. What is Selenium WebDriver?

Answer: Selenium WebDriver is an automation API used to control web browsers and interact with web elements.

Q3. What is an assertion?

Answer: An assertion compares an expected condition with the actual result and indicates whether the condition is satisfied. Example: assert 'inventory' in driver.current_url.

Q4. What is exception handling?

Answer: Exception handling catches and manages runtime errors so the program can report problems instead of stopping unexpectedly.

Q5. Which exception can occur when an element cannot be found?

Answer: NoSuchElementException.

16. Final Checklist

☐ Create and run tutorial15.py in PyCharm.

☐ Verify all five scenarios execute as expected.

☐ Check that the final console output shows the actual status of each test.

☐ Include screenshots of the login page, successful login, invalid password, invalid username, empty credentials, logout, source code, and console output.

☐ Fill in the enrollment number on the cover page before submission.